A natural transformation is an isomorphism precisely when all its components are

The Univalent Foundations Program

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Editorial note. The counted statement is the author's 'ct:natiso' (source lines 335-339, Lemma 9.2.4); the proof environment starts on the next line, so the pairing is by adjacency.

Editorial note. No context span is retained: the statement and proof are self-contained and use no label from outside the retained range.

Editorial note. The retained counter values (chapter 9, section 2, theorem offset 3) make the counted lemma print as Lemma 9.2.4, matching the original book numbering taken from the label-number dump of the pinned snapshot.

Editorial note. The retained preamble is the chapter's own retained starter preamble for this pinned source.

Editorial scope. One original result of the HoTT Book, counted once: natural isomorphisms are the componentwise isomorphisms (Lemma 9.2.4).

Allowed standard prerequisites (editorial).

- Functor precategory B^A. The precategory B^A has functors A -> B as objects and natural transformations as morphisms, with composition and identity defined componentwise.. It is the precategory in which the isomorphism is taken.
- Isomorphism in a precategory. An isomorphism is a morphism with a two-sided inverse.. It is the notion compared componentwise with isomorphisms in B.

Lemma 9.2.4. *A natural transformation $\gamma : F \to G$ is an isomorphism in B^A if and only if each γ_a is an isomorphism in B.*

Proof. If γ is an isomorphism, then we have $\delta : G \to F$ that is its inverse. By definition of composition in B^A, $(\delta\gamma)_a \equiv \delta_a\gamma_a$ and similarly $(\gamma\delta)_a \equiv \gamma_a\delta_a$. Thus, $\delta\gamma = 1_F$ and $\gamma\delta = 1_G$ imply $\delta_a\gamma_a = 1_{Fa}$ and $\gamma_a\delta_a = 1_{Ga}$, so γ_a is an isomorphism.

Conversely, suppose each γ_a is an isomorphism, with inverse called δ_a, say. We define a natural transformation $\delta : G \to F$ with components δ_a; for the naturality axiom we have

$$Ff \circ \delta_a = \delta_b \circ \gamma_b \circ Ff \circ \delta_a = \delta_b \circ Gf \circ \gamma_a \circ \delta_a = \delta_b \circ Gf.$$

Now since composition and identity of natural transformations is determined on their components, we have $\gamma\delta = 1_G$ and $\delta\gamma = 1_F$. $\square$